Thứ ngày tháng năm

P_1: 2.0 Dương Khánh Thiện ①

P_2: 2.5 N25DCCE008

P_3: 2.5 Problem 1: Singly Linked List - In Place Reserval & Pointer

P_4: 2.25 tracy

9.25 Given a singly linked list: head → [10] → [25] → [40] → [15] → Null

2.00 Task 1: Write a function/algorithm in Pseudocode to reverse the singly linked list, including three pointers (Prev, curr, next) to complete your function

Task 2 Draw a diagram or step-by-step pointer trace table showing the positions and values of the three pointers (Prev, curr, next) after each iteration until the list is fully reversed.

~~Problem 2:~~ (Solve)

1.25 Task 1:

```
Function ReverseList(Head):
    Prev = null
    curr = Head
    While(curr is not null)
        next = curr.next
        curr.next = prev
```

prev ← NULL

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curr = next

head = prev

return head

0,75 Task 2

Linked Structure ?

Iteration	Prev	curr	next
Initial	Null	[10]	Null
1	[10]	[25]	[25]
2	[25]	[40]	[40]
3	[40]	[15]	[15]
4	[15]	Null	Null

Final result?

2,5 Problem 2 Circular linked list - Insert a Head

Given a Singly circular linked list

Head -> [10] -> [20] -> [30] => (Point back to Head)

Task 1 Write pseudocode for the function InsertHead (Head, newValue) to insert new node containing new value at the front of the circular linked list while maintaining its circular structure

Task 2 Trace the execution steps (Use text description or Pointer diagrams) When inserting new Value = 5 into to given list

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Solve

1,25 Task 1

```
Function InsertHead(Head, NewValue):
    newNode = new Node(newValue)
    if (Head is Null)
        newNode.next = newNode
        return newNode
    newNode.next = Head
    temp = Head
    while (temp.next != Head)
        temp = temp.next
    temp.next = newNode
    Head = newNode
    Head = newNode
    return Head
```

1,25 Task 2

1 newNode = new Node(~~Node 5 -> 70~~)

2 newNode.next = Head (Node 5 -> 70)

3 Traverse temp pointer to the last node (node 70)

4 Assign temp.next = new Node (point 30 to 5)

5 update Head = new Node (New ~~Head~~ Head is now 5)

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9.5

Problem 3: Doubly Linked List - Insert before Target & Edge Cases

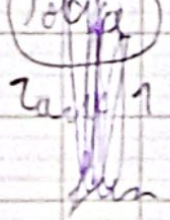
Given a doubly linked list containing integers

Task 1: Write pseudo code for Insert Before (head, target, new value) to insert a new node containing new value immediately before the first occurrence of target

Task 2: List the exact order of pointer assignments required and briefly justify why this specific sequence avoids memory leaks or broken links

Task 3: Explicitly handle the following two edge cases in code / pseudo code:

1. Target is located at the head node (head).
2. Target does not exist in the list



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Dương Hữu Thiện (2)
N25DECE068
9,5 (Solve) Problem 3

1,75 Task 1 & 3:

function InsertBefore(Head, target, newValue)
    # If Head is Null: return Null // Edge Case 2
target does not exist.
    newNode = newNode(newValue)
    if (Head.data == target: // Edge Case 1: Target
is at the Head)
        newNode.next = Head
        Head.prev = newNode
        Head = newNode
        return Head
    curr = Head
    While curr is not Null and curr.data != target
        curr = curr.next
    If curr is Null: return Head // Edge case 2: Target
not found
        newNode.next = curr

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new Node.prev = curr.prev

curr.prev.next = newNode

~~curr~~ curr.prev = newNode

~~return~~ return head

0,75 Task 2

1 newNode.next = curr

2 ~~new~~ newNode.prev = curr.prev

3 curr.prev.next = newNode

4 curr.prev = newNode

2,25 Problem 4 Sorted Linked list - In-Order Insertion

Given an Singly ~~Sorted~~ Linked list sorted in ascending order: head → [2] → [5] → [8] → Null

Task 1: ~~Hãy~~ Write the function SortedInsert (Head, val) to insert val into the list such that the ascending order is preserved

Task 2: Construct a Step-by-step trace table declaring pointer iteration and condition evaluations when inserting val: 6 into the list

Solve

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1,25 Task 1

```
function SortedInsert(Head, val):
    newNode = newNode(val)
    if (Head is null or Head.data >= val.)
        newNode.next = Head
        Head = newNode
        return Head
    curr = Head
    while (curr.next is not null and curr.
    curr.next.data < val):
        curr = curr.next
    newNode.next = curr.next
    curr.next = newNode
    return Head
```

1,00 Task 2

Step	Condition (curr.next != null)
init	
i+1th	
i th 2	
insert	

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Task 1

Step	Condition (curr_next != null && curr_next.data < val)	curr position	Action
start	N/A	[2]	Start loop
iter 1	True (5 < 6)	move to [5]	Continue loop
iter 2	False (8 < 6 is False)	Stay at [5]	Exit loop
insert	N/A	At [5]	newNode(6)
			next = [5].
			next = newNode

Final result: ?

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